

The Two Hills of AI: Automation vs. Invention

Most enterprise AI strategy today is framed around productivity: take an existing manual process, automate it, count the hours saved. It's measurable, it's safe, and it's real value. I won't dismiss it.

But here's the uncomfortable truth: **productivity gains from AI automation are table stakes**. If you can automate a process with AI, so can your competitor — usually by buying the same tools you did. **Efficiency that everyone can purchase is not a differentiator**. It's the new baseline.

The more interesting question is: **what can AI let us do that was previously impossible?**

That's the difference between **automating existing processes** and **inventing new capabilities**. And in my experience, the second path - done right - delivers the productivity gains anyway, as a byproduct. Invention subsumes automation. The reverse is not true.

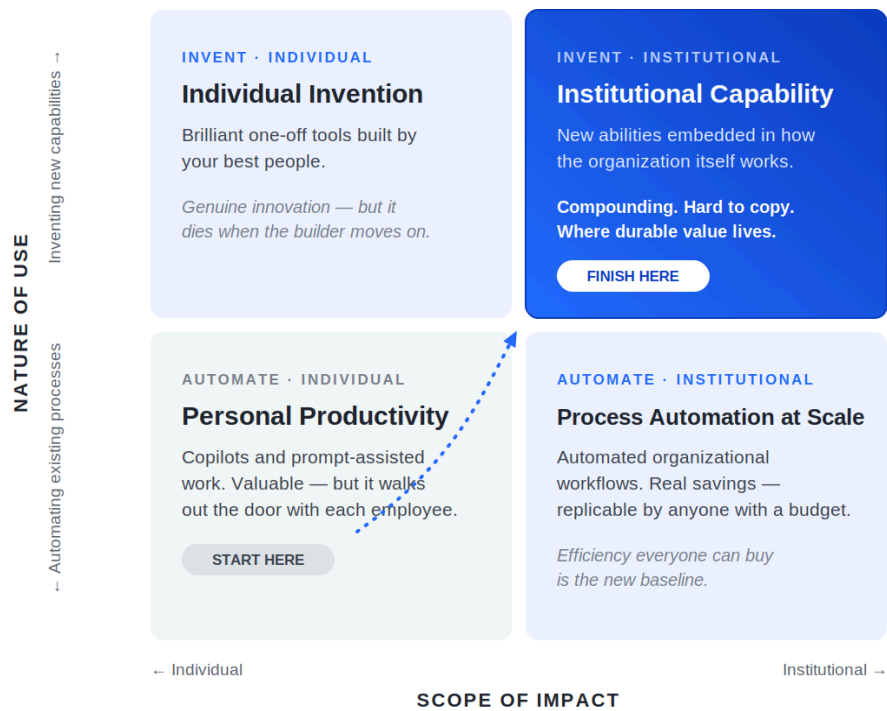
A framework for deciding where to invest

Simple 2x2 to think about AI investments:

- X-axis: Scope of impact - from **individual** to **institutional**
- Y-axis: Nature of use - from **automating** existing processes to **inventing** new capabilities

The Two Hills of AI

Automating what exists vs. inventing what was impossible — where durable value lives



Invention subsumes automation. The reverse is not true.

Every AI journey starts in the bottom half — that's where the quick wins are. But durable value lives in one quadrant only: **institutional capability**. New abilities embedded in how the organization itself works. Compounding. Hard to copy.

Here's what that looked like in practice, from work I've been close to in enterprise quality engineering.

When we set out to reimagine end-to-end UI test automation at scale, the obvious path was clear: use AI to generate test scripts from natural language. Most people expected it. It would have been a legitimate productivity win.

But when we looked at the data across real-world usage, the limitations of that approach became clear fast: generated scripts couldn't keep pace with how quickly the underlying application changed. We'd have been automating a process whose fundamental constraints remained — just producing output faster, not better.

So we asked a different question: **where should the intelligence actually live?** Our answer: in understanding the intent of a use case and figuring out execution the way a human tester would — not in generating scripts.

That question led to a platform built around reasoning about user intent rather than producing fixed scripts. Because we built for capability rather than a specific tool, the same approach ended up covering multiple platforms and device types — something script generation alone could never have given us.

The outcomes tell the story: the platform scales to a very large number of use cases and keeps its own tests current as the product evolves. And the productivity gains followed as a byproduct, with the compounding value still ahead of us.

We didn't choose between innovation and productivity. **We chose innovation, and productivity came with it.**

So here's my question for every AI initiative on your roadmap:

Is this automating something we already do — or inventing something we couldn't do before?

Both are legitimate. Only one compounds.

Personal productivity is a fine place to start. Institutional value is the only place worth finishing.